Ch07_nonlin_lab

April 28, 2026

[1]:
```
from functools import partial
print = partial(print, flush=True)
```

[2]:
```
import numpy as np, pandas as pd
from matplotlib.pyplot import subplots
import statsmodels.api as sm
from ISLP import load_data
from ISLP.models import (summarize,
                         poly,
                         ModelSpec as MS)
from statsmodels.stats.anova import anova_lm
```

[3]:
```
from pygam import (s as s_gam,
                   l as l_gam,
                   f as f_gam,
                   LinearGAM,
                   LogisticGAM)

from ISLP.transforms import (BSpline,
                             NaturalSpline)
from ISLP.models import bs, ns
from ISLP.pygam import (approx_lam,
                        degrees_of_freedom,
                        plot as plot_gam,
                        anova as anova_gam)
```

[4]:
```
Wage = load_data('Wage')
y = Wage['wage']
age = Wage['age']
```

[5]:
```
poly_age = MS([poly('age', degree=4)]).fit(Wage)
M = sm.OLS(y, poly_age.transform(Wage)).fit()
summarize(M)
```

[5]:

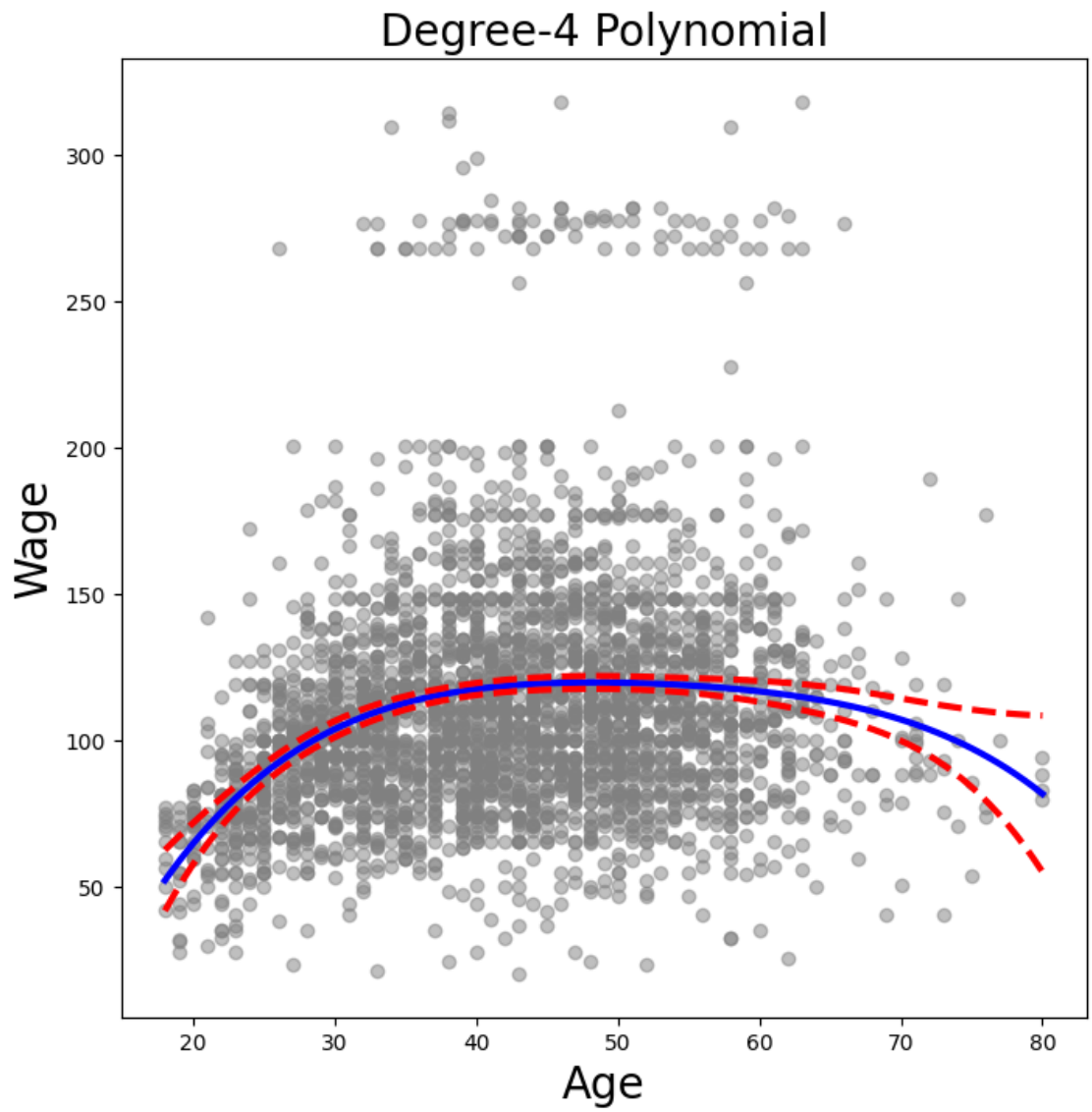
	coef	std err	t	P>|t|
intercept	111.7036	0.729	153.283	0.000
poly(age, degree=4)[0]	447.0679	39.915	11.201	0.000
poly(age, degree=4)[1]	-478.3158	39.915	-11.983	0.000

```
poly(age, degree=4)[2]  125.5217   39.915    3.145  0.002
poly(age, degree=4)[3]  -77.9112   39.915   -1.952  0.051
```

[6]:
```
age_grid = np.linspace(age.min(),
                       age.max(),
                       100)
age_df = pd.DataFrame({'age': age_grid})
```

[7]:
```
def plot_wage_fit(age_df,
                  basis,
                  title):

    X = basis.transform(Wage)
    Xnew = basis.transform(age_df)
    M = sm.OLS(y, X).fit()
    preds = M.get_prediction(Xnew)
    bands = preds.conf_int(alpha=0.05)
    fig, ax = subplots(figsize=(8,8))
    ax.scatter(age,
               y,
               facecolor='gray',
               alpha=0.5)
    for val, ls in zip([preds.predicted_mean,
                        bands[:,0],
                        bands[:,1]],
                       ['b','r--','r--']):
        ax.plot(age_df.values, val, ls, linewidth=3)
    ax.set_title(title, fontsize=20)
    ax.set_xlabel('Age', fontsize=20)
    ax.set_ylabel('Wage', fontsize=20);
    return ax
```

[8]:
```
plot_wage_fit(age_df,
              poly_age,
              'Degree-4 Polynomial');
```



[9]:
```
models = [MS([poly('age', degree=d)])
          for d in range(1, 6)]
Xs = [model.fit_transform(Wage) for model in models]
anova_lm(*[sm.OLS(y, X_).fit()
           for X_ in Xs])
```

[9]:

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	2998.0	5.022216e+06	0.0	NaN	NaN	NaN
1	2997.0	4.793430e+06	1.0	228786.010128	143.593107	2.363850e-32
2	2996.0	4.777674e+06	1.0	15755.693664	9.888756	1.679202e-03
3	2995.0	4.771604e+06	1.0	6070.152124	3.809813	5.104620e-02
4	2994.0	4.770322e+06	1.0	1282.563017	0.804976	3.696820e-01

[10]:
```
summarize(M)
```

[10]:

	coef	std err	t	P>|t|
intercept	111.7036	0.729	153.283	0.000
poly(age, degree=4)[0]	447.0679	39.915	11.201	0.000
poly(age, degree=4)[1]	-478.3158	39.915	-11.983	0.000
poly(age, degree=4)[2]	125.5217	39.915	3.145	0.002
poly(age, degree=4)[3]	-77.9112	39.915	-1.952	0.051

[11]:
```
(-11.983)**2
```

[11]: 143.59228900000002

[12]:
```
models = [MS(['education', poly('age', degree=d)])
          for d in range(1, 4)]
XEs = [model.fit_transform(Wage)
       for model in models]
anova_lm(*[sm.OLS(y, X_).fit() for X_ in XEs])
```

[12]:

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	2997.0	3.902335e+06	0.0	NaN	NaN	NaN
1	2996.0	3.759472e+06	1.0	142862.701185	113.991883	3.838075e-26
2	2995.0	3.753546e+06	1.0	5926.207070	4.728593	2.974318e-02

[13]:
```
X = poly_age.transform(Wage)
high_earn = Wage['high_earn'] = y > 250 # shorthand
glm = sm.GLM(y > 250,
             X,
             family=sm.families.Binomial())
B = glm.fit()
summarize(B)
```

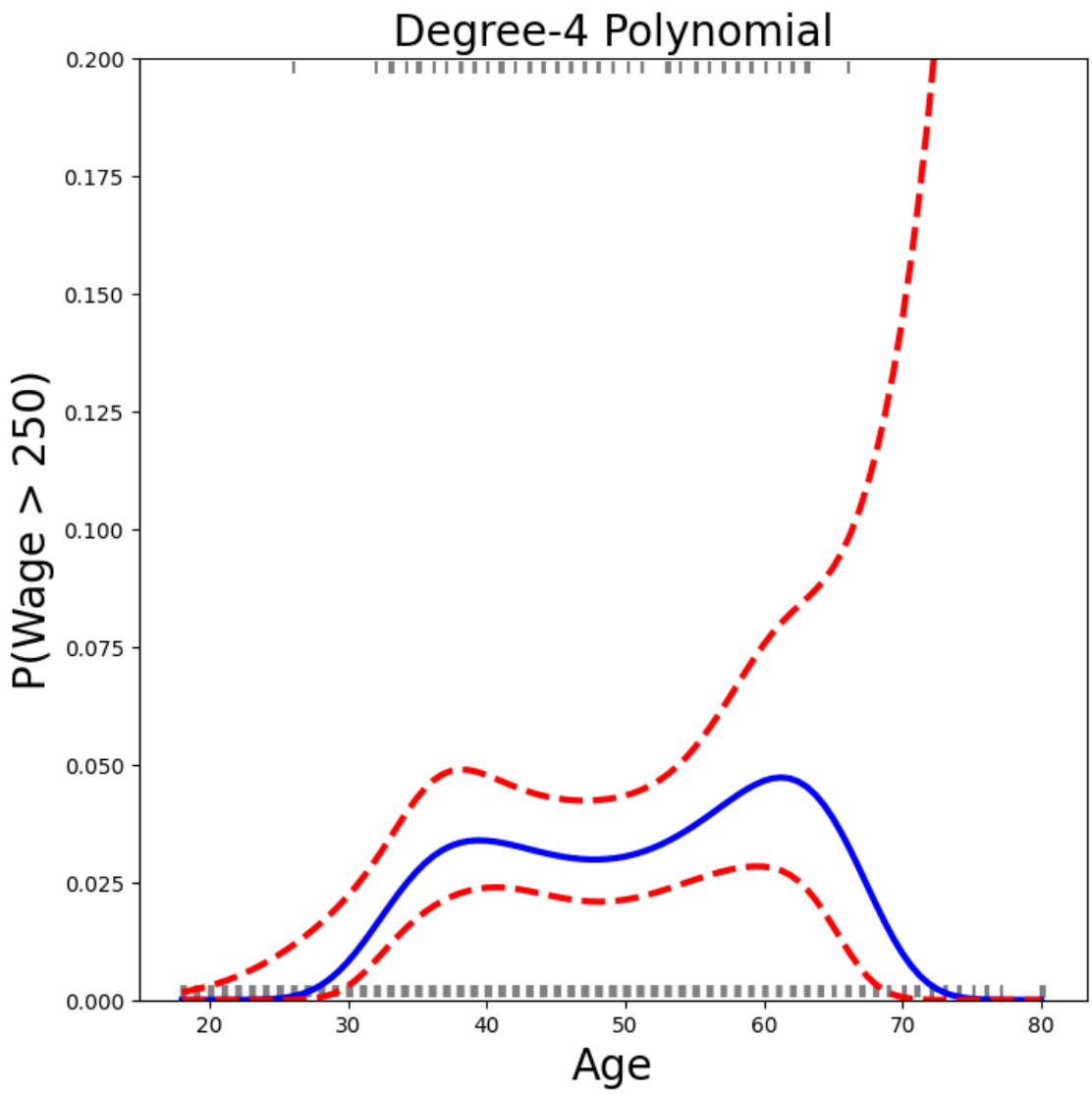
[13]:

	coef	std err	z	P>|z|
intercept	-4.3012	0.345	-12.457	0.000
poly(age, degree=4)[0]	71.9642	26.133	2.754	0.006
poly(age, degree=4)[1]	-85.7729	35.929	-2.387	0.017
poly(age, degree=4)[2]	34.1626	19.697	1.734	0.083
poly(age, degree=4)[3]	-47.4008	24.105	-1.966	0.049

[14]:
```
newX = poly_age.transform(age_df)
preds = B.get_prediction(newX)
bands = preds.conf_int(alpha=0.05)
```

[15]:
```
fig, ax = subplots(figsize=(8,8))
rng = np.random.default_rng(0)
ax.scatter(age +
           0.2 * rng.uniform(size=y.shape[0]),
           np.where(high_earn, 0.198, 0.002),
```

```
            fc='gray',
            marker='|')
for val, ls in zip([preds.predicted_mean,
                    bands[:,0],
                    bands[:,1]],
                   ['b','r--','r--']):
    ax.plot(age_df.values, val, ls, linewidth=3)
ax.set_title('Degree-4 Polynomial', fontsize=20)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylim([0,0.2])
ax.set_ylabel('P(Wage > 250)', fontsize=20);
```



[16]:
```
cut_age = pd.qcut(age, 4)
summarize(sm.OLS(y, pd.get_dummies(cut_age)).fit())
```

[16]:
```
                     coef  std err       t  P>|t|
(17.999, 33.75]   94.1584    1.478  63.692    0.0
(33.75, 42.0]    116.6608    1.470  79.385    0.0
(42.0, 51.0]     119.1887    1.416  84.147    0.0
(51.0, 80.0]     116.5717    1.559  74.751    0.0
```

[17]:
```
bs_ = BSpline(internal_knots=[25,40,60], intercept=True).fit(age)
bs_age = bs_.transform(age)
bs_age.shape
```

[17]:
```
(3000, 7)
```

[18]:
```
bs_age = MS([bs('age', internal_knots=[25,40,60])])
Xbs = bs_age.fit_transform(Wage)
M = sm.OLS(y, Xbs).fit()
summarize(M)
```

[18]:
```
                                          coef  std err      t  P>|t|
intercept                                60.4937    9.460  6.394  0.000
bs(age, internal_knots=[25, 40, 60])[0]   3.9805   12.538  0.317  0.751
bs(age, internal_knots=[25, 40, 60])[1]  44.6310    9.626  4.636  0.000
bs(age, internal_knots=[25, 40, 60])[2]  62.8388   10.755  5.843  0.000
bs(age, internal_knots=[25, 40, 60])[3]  55.9908   10.706  5.230  0.000
bs(age, internal_knots=[25, 40, 60])[4]  50.6881   14.402  3.520  0.000
bs(age, internal_knots=[25, 40, 60])[5]  16.6061   19.126  0.868  0.385
```

[19]:
```
bs_age = MS([bs('age',
                internal_knots=[25,40,60],
                name='bs(age)')])
Xbs = bs_age.fit_transform(Wage)
M = sm.OLS(y, Xbs).fit()
summarize(M)
```

[19]:
```
               coef  std err      t  P>|t|
intercept   60.4937    9.460  6.394  0.000
bs(age)[0]   3.9805   12.538  0.317  0.751
bs(age)[1]  44.6310    9.626  4.636  0.000
bs(age)[2]  62.8388   10.755  5.843  0.000
bs(age)[3]  55.9908   10.706  5.230  0.000
bs(age)[4]  50.6881   14.402  3.520  0.000
bs(age)[5]  16.6061   19.126  0.868  0.385
```

[20]:
```
BSpline(df=6).fit(age).internal_knots_
```

[20]:
```
array([33.75, 42.  , 51.  ])
```

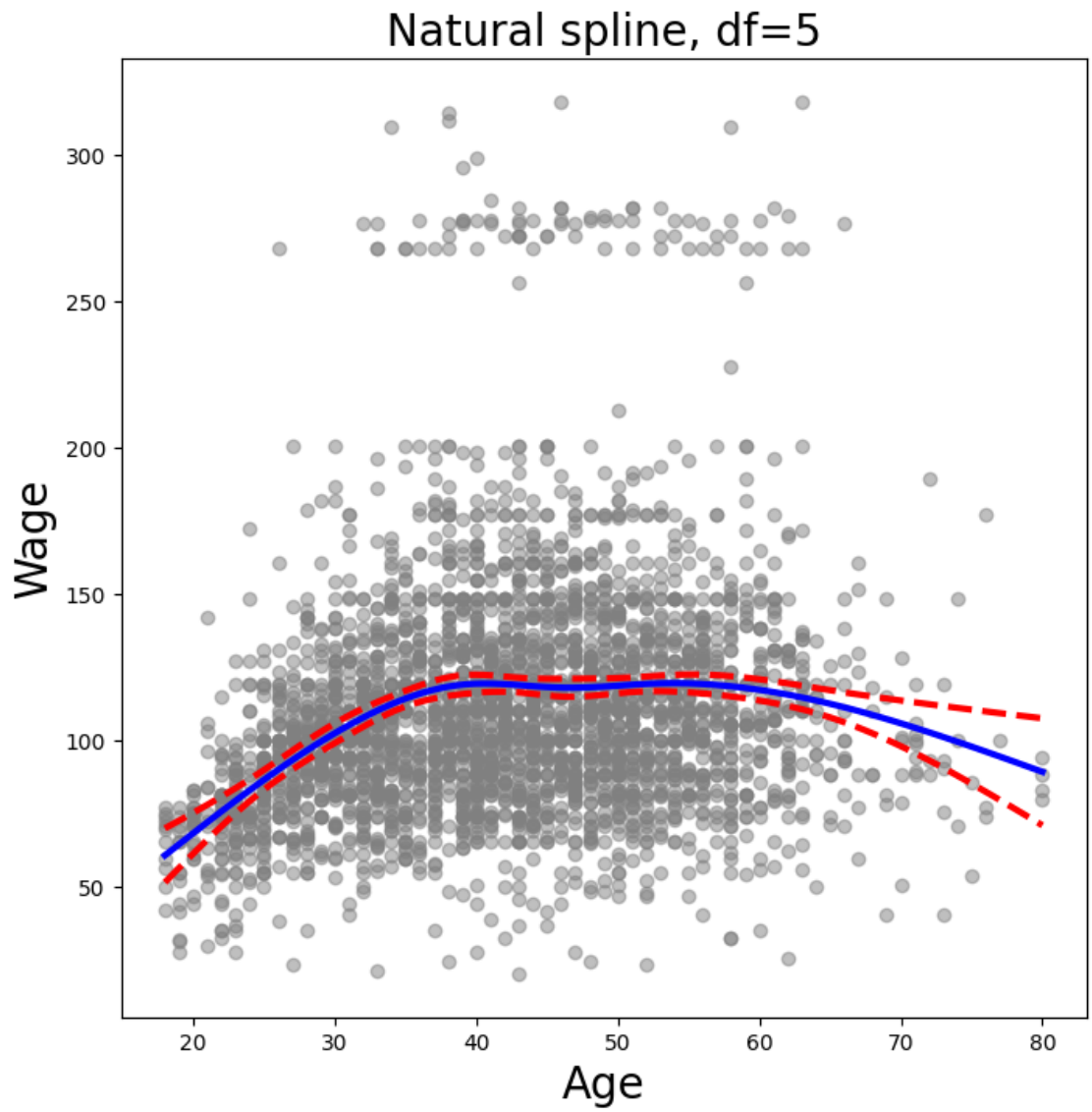
[21]:
```
bs_age0 = MS([bs('age',
                 df=3,
                 degree=0)]).fit(Wage)
Xbs0 = bs_age0.transform(Wage)
summarize(sm.OLS(y, Xbs0).fit())
```

[21]:
```
                              coef  std err       t  P>|t|
intercept                   94.1584    1.478  63.687    0.0
bs(age, df=3, degree=0)[0]  22.3490    2.152  10.388    0.0
bs(age, df=3, degree=0)[1]  24.8076    2.044  12.137    0.0
bs(age, df=3, degree=0)[2]  22.7814    2.087  10.917    0.0
```

[22]:
```
ns_age = MS([ns('age', df=5)]).fit(Wage)
M_ns = sm.OLS(y, ns_age.transform(Wage)).fit()
summarize(M_ns)
```

[22]:
```
                      coef  std err       t  P>|t|
intercept         60.4752    4.708  12.844  0.000
ns(age, df=5)[0]  61.5267    4.709  13.065  0.000
ns(age, df=5)[1]  55.6912    5.717   9.741  0.000
ns(age, df=5)[2]  46.8184    4.948   9.463  0.000
ns(age, df=5)[3]  83.2036   11.918   6.982  0.000
ns(age, df=5)[4]   6.8770    9.484   0.725  0.468
```

[23]:
```
plot_wage_fit(age_df,
              ns_age,
              'Natural spline, df=5');
```



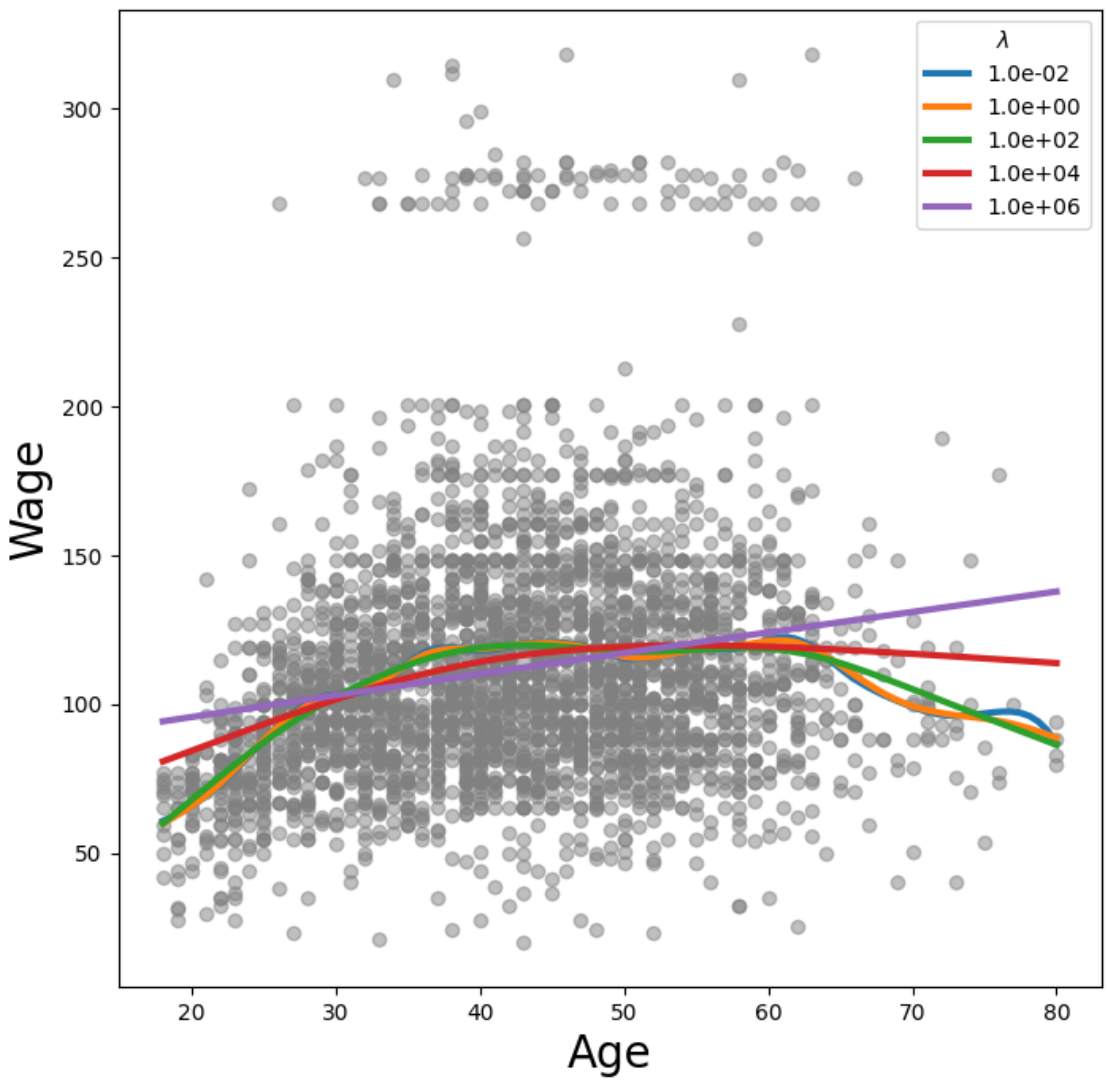
[24]:
```
X_age = np.asarray(age).reshape((-1,1))
gam = LinearGAM(s_gam(0, lam=0.6))
gam.fit(X_age, y)
```

[24]:
```
LinearGAM(callbacks=[Deviance(), Diffs()], fit_intercept=True,
   max_iter=100, scale=None, terms=s(0) + intercept, tol=0.0001,
   verbose=False)
```

[25]:
```
fig, ax = subplots(figsize=(8,8))
ax.scatter(age, y, facecolor='gray', alpha=0.5)
for lam in np.logspace(-2, 6, 5):
    gam = LinearGAM(s_gam(0, lam=lam)).fit(X_age, y)
```

```
    ax.plot(age_grid,
            gam.predict(age_grid),
            label='{:.1e}'.format(lam),
            linewidth=3)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylabel('Wage', fontsize=20);
ax.legend(title='$\lambda$');
```

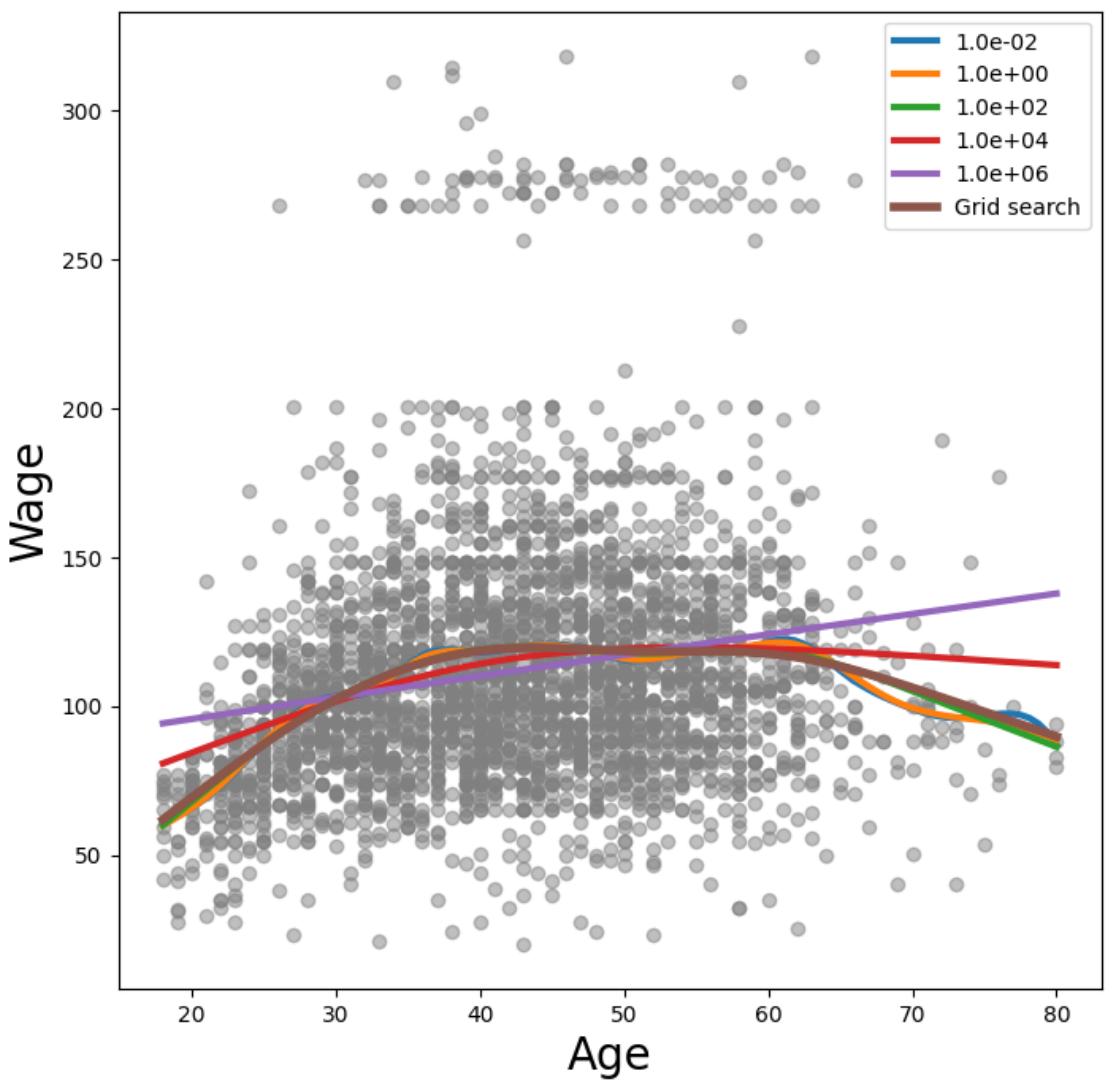
```
<>:11: SyntaxWarning: "\l" is an invalid escape sequence. Such sequences will
not work in the future. Did you mean "\\l"? A raw string is also an option.
<>:11: SyntaxWarning: "\l" is an invalid escape sequence. Such sequences will
not work in the future. Did you mean "\\l"? A raw string is also an option.
C:\Users\archa\AppData\Local\Temp\ipykernel_30340\1121239147.py:11:
SyntaxWarning: "\l" is an invalid escape sequence. Such sequences will not work
in the future. Did you mean "\\l"? A raw string is also an option.
  ax.legend(title='$\lambda$');
```



[26]:
```
gam_opt = gam.gridsearch(X_age, y)
ax.plot(age_grid,
        gam_opt.predict(age_grid),
        label='Grid search',
        linewidth=4)
ax.legend()
fig
```

```
  0% (0 of 11) |                          | Elapsed Time: 0:00:00 ETA:  --:--:--
 27% (3 of 11) |######                   | Elapsed Time: 0:00:00 ETA:   0:00:00
 54% (6 of 11) |#############            | Elapsed Time: 0:00:00 ETA:   0:00:00
 81% (9 of 11) |####################     | Elapsed Time: 0:00:00 ETA:   0:00:00
100% (11 of 11) |########################| Elapsed Time: 0:00:00 Time:  0:00:00
```

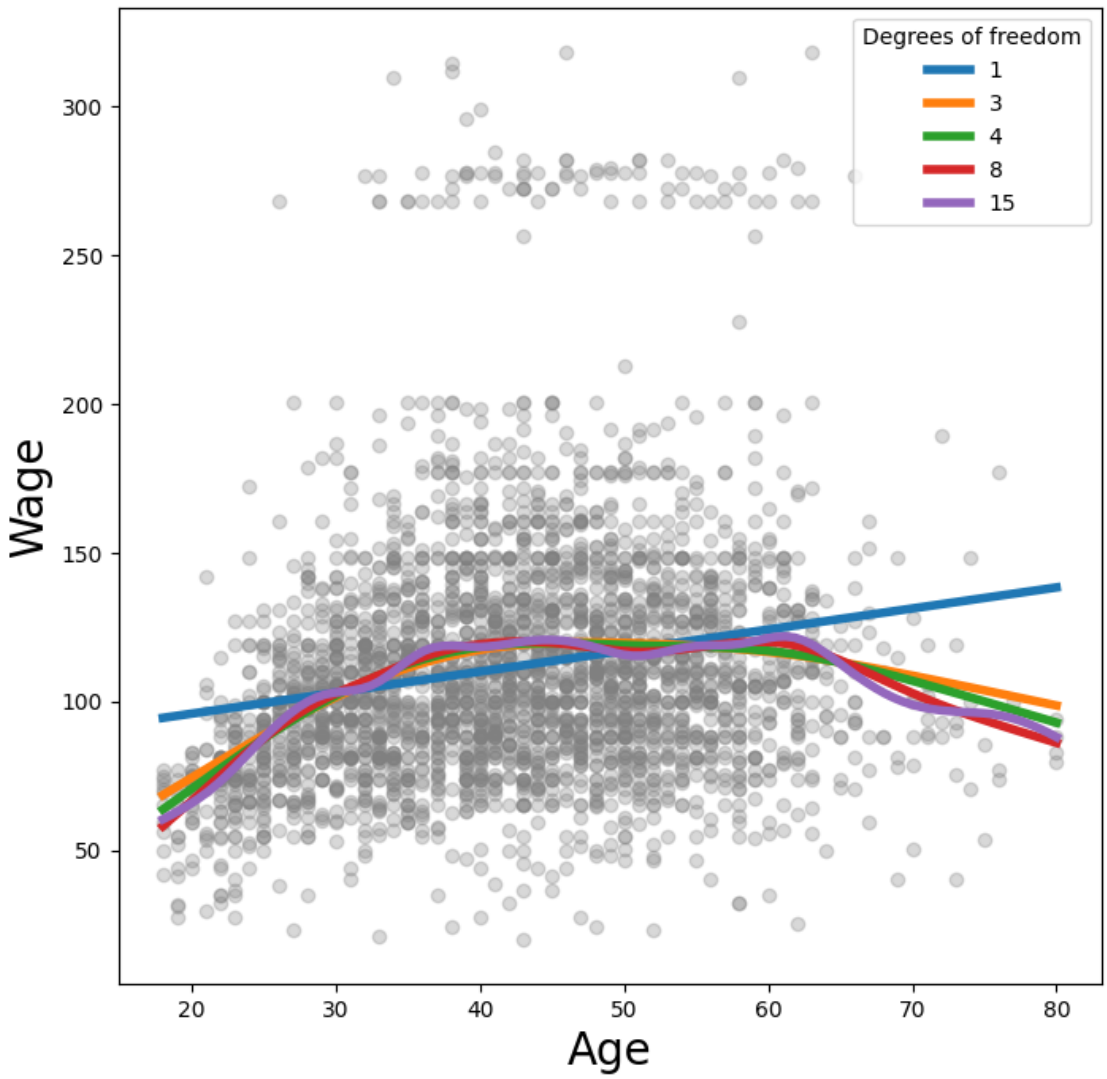
[26]:



[27]:
```
age_term = gam.terms[0]
lam_4 = approx_lam(X_age, age_term, 4)
age_term.lam = lam_4
degrees_of_freedom(X_age, age_term)
```

[27]:
```
np.float64(4.000000100000307)
```

[28]:
```
fig, ax = subplots(figsize=(8,8))
ax.scatter(X_age,
           y,
           facecolor='gray',
           alpha=0.3)
for df in [1,3,4,8,15]:
    lam = approx_lam(X_age, age_term, df+1)
    age_term.lam = lam
    gam.fit(X_age, y)
    ax.plot(age_grid,
            gam.predict(age_grid),
            label='{:d}'.format(df),
            linewidth=4)
ax.set_xlabel('Age', fontsize=20)
ax.set_ylabel('Wage', fontsize=20);
ax.legend(title='Degrees of freedom');
```

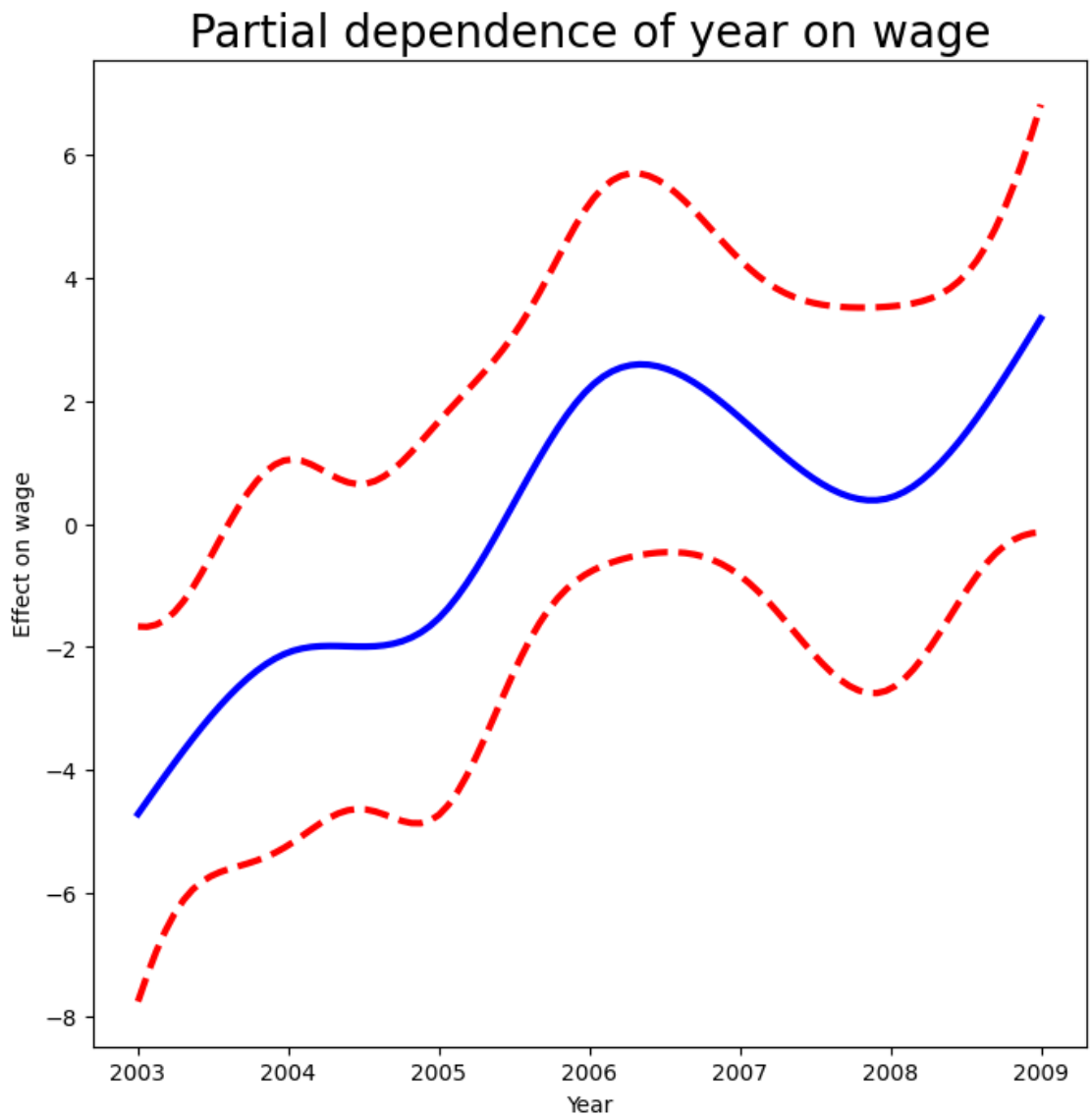


[29]:
```
ns_age = NaturalSpline(df=4).fit(age)
ns_year = NaturalSpline(df=5).fit(Wage['year'])
Xs = [ns_age.transform(age),
      ns_year.transform(Wage['year']),
      pd.get_dummies(Wage['education']).values]
X_bh = np.hstack(Xs)
gam_bh = sm.OLS(y, X_bh).fit()
```

[30]:
```
age_grid = np.linspace(age.min(),
                       age.max(),
                       100)
X_age_bh = X_bh.copy()[:100]
X_age_bh[:] = X_bh[:].mean(0)[None,:]
```

```
X_age_bh[:,:4] = ns_age.transform(age_grid)
preds = gam_bh.get_prediction(X_age_bh)
bounds_age = preds.conf_int(alpha=0.05)
partial_age = preds.predicted_mean
center = partial_age.mean()
partial_age -= center
bounds_age -= center
fig, ax = subplots(figsize=(8,8))
ax.plot(age_grid, partial_age, 'b', linewidth=3)
ax.plot(age_grid, bounds_age[:,0], 'r--', linewidth=3)
ax.plot(age_grid, bounds_age[:,1], 'r--', linewidth=3)
ax.set_xlabel('Age')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of age on wage', fontsize=20);
```

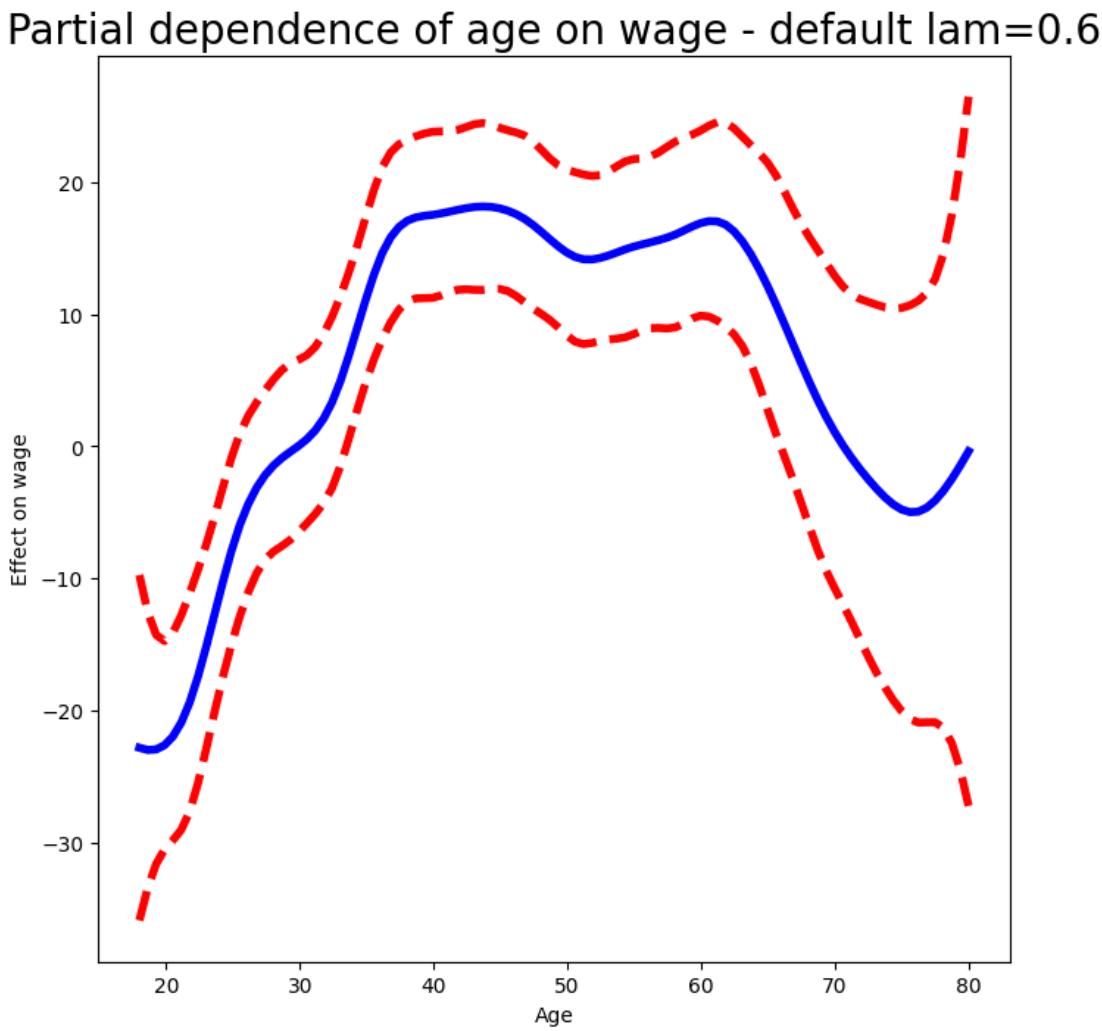
[31]:
```
year_grid = np.linspace(2003, 2009, 100)
year_grid = np.linspace(Wage['year'].min(),
                        Wage['year'].max(),
                        100)
X_year_bh = X_bh.copy()[:100]
X_year_bh[:] = X_bh[:].mean(0)[None,:]
X_year_bh[:,4:9] = ns_year.transform(year_grid)
preds = gam_bh.get_prediction(X_year_bh)
bounds_year = preds.conf_int(alpha=0.05)
partial_year = preds.predicted_mean
center = partial_year.mean()
partial_year -= center
bounds_year -= center
fig, ax = subplots(figsize=(8,8))
ax.plot(year_grid, partial_year, 'b', linewidth=3)
ax.plot(year_grid, bounds_year[:,0], 'r--', linewidth=3)
ax.plot(year_grid, bounds_year[:,1], 'r--', linewidth=3)
ax.set_xlabel('Year')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of year on wage', fontsize=20);
```



[32]:
```
gam_full = LinearGAM(s_gam(0) +
                     s_gam(1, n_splines=7) +
                     f_gam(2, lam=0))
Xgam = np.column_stack([age,
                        Wage['year'],
                        Wage['education'].cat.codes])
gam_full = gam_full.fit(Xgam, y)
```

[33]:
```
fig, ax = subplots(figsize=(8,8))
plot_gam(gam_full, 0, ax=ax)
ax.set_xlabel('Age')
ax.set_ylabel('Effect on wage')
```

```
ax.set_title('Partial dependence of age on wage - default lam=0.6',␣
 ↪fontsize=20);
```

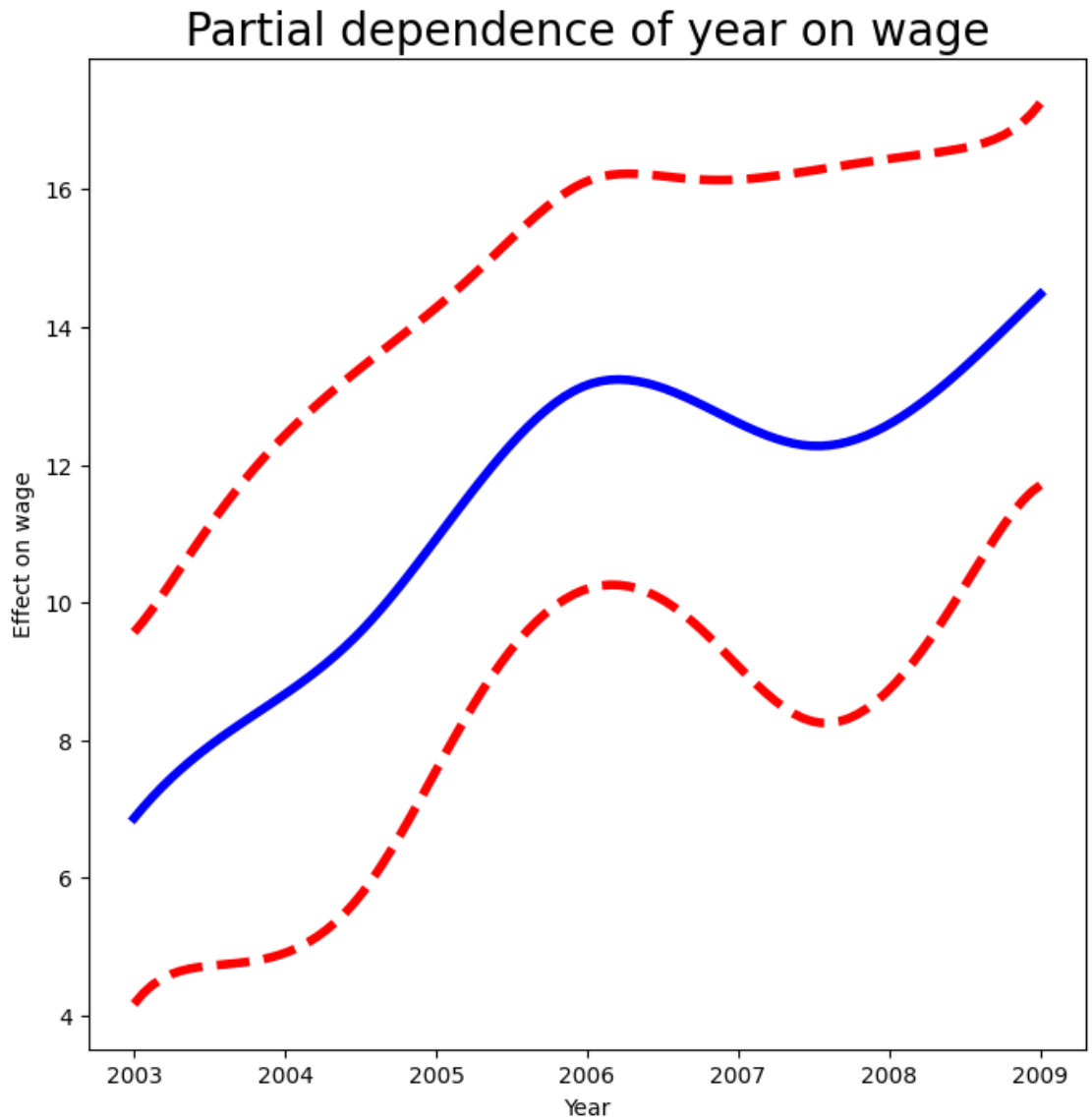


[34]:
```
age_term = gam_full.terms[0]
age_term.lam = approx_lam(Xgam, age_term, df=4+1)
year_term = gam_full.terms[1]
year_term.lam = approx_lam(Xgam, year_term, df=4+1)
gam_full = gam_full.fit(Xgam, y)
```

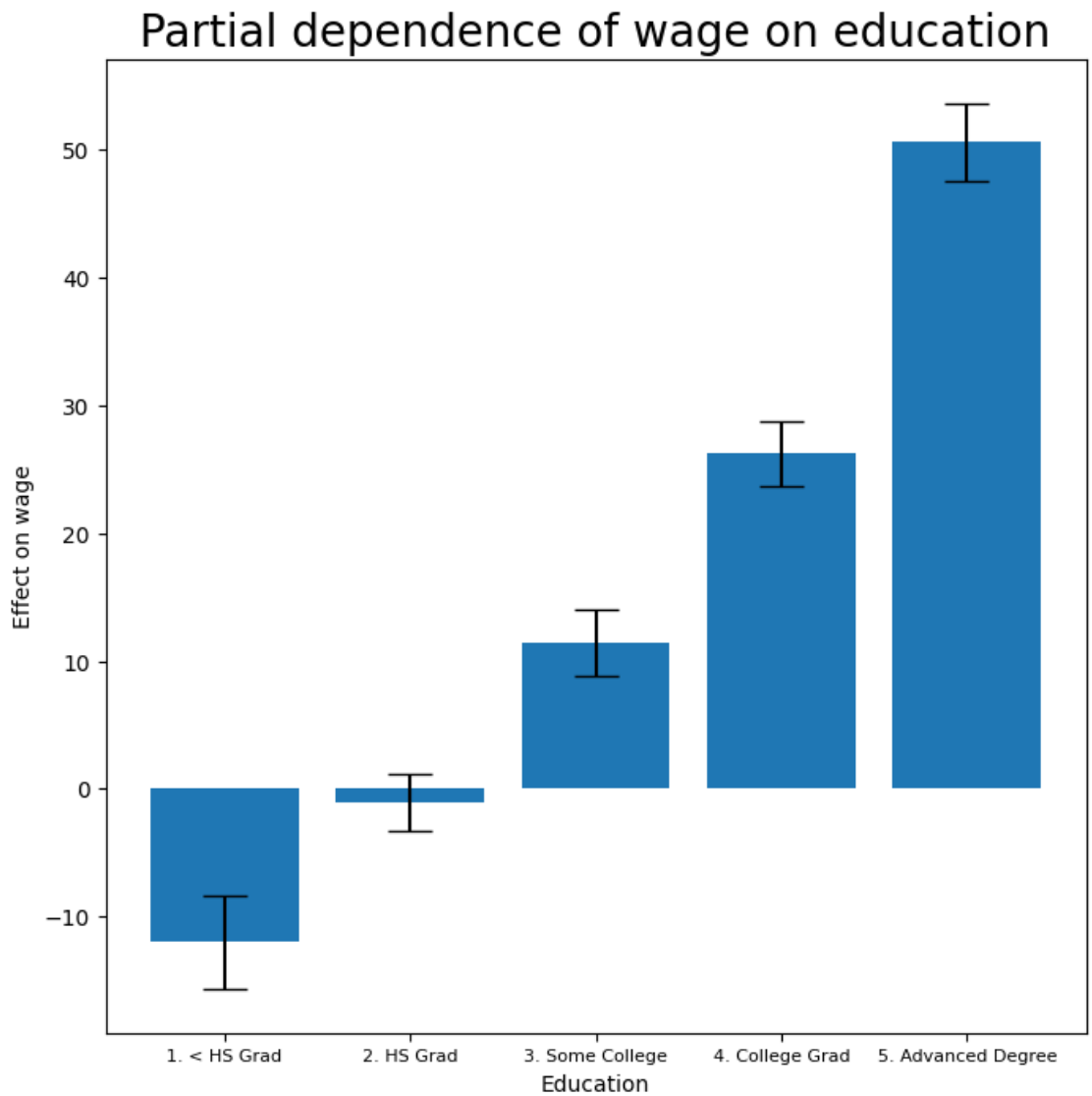
[35]:
```
fig, ax = subplots(figsize=(8,8))
plot_gam(gam_full,
         1,
         ax=ax)
ax.set_xlabel('Year')
ax.set_ylabel('Effect on wage')
```

```
ax.set_title('Partial dependence of year on wage', fontsize=20)
```

[35]: Text(0.5, 1.0, 'Partial dependence of year on wage')



[36]:
```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_full, 2)
ax.set_xlabel('Education')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of wage on education',
             fontsize=20);
ax.set_xticklabels(Wage['education'].cat.categories, fontsize=8);
```



[37]:
```
gam_0 = LinearGAM(age_term + f_gam(2, lam=0))
gam_0.fit(Xgam, y)
gam_linear = LinearGAM(age_term +
                       l_gam(1, lam=0) +
                       f_gam(2, lam=0))
gam_linear.fit(Xgam, y)
```

[37]:
```
LinearGAM(callbacks=[Deviance(), Diffs()], fit_intercept=True,
   max_iter=100, scale=None, terms=s(0) + l(1) + f(2) + intercept,
   tol=0.0001, verbose=False)
```

[38]:
```
anova_gam(gam_0, gam_linear, gam_full)
```

[38]:

	deviance	df	deviance_diff	df_diff	F	pvalue
0	105402.432206	2991.004005	NaN	NaN	NaN	NaN
1	105134.624170	2990.005190	267.808037	0.998815	7.625329	0.016210
2	105030.684668	2987.007254	103.939502	2.997936	0.986003	0.429637

[39]:

```
gam_0 = LinearGAM(year_term +
                  f_gam(2, lam=0))
gam_linear = LinearGAM(l_gam(0, lam=0) +
                       year_term +
                       f_gam(2, lam=0))
gam_0.fit(Xgam, y)
gam_linear.fit(Xgam, y)
anova_gam(gam_0, gam_linear, gam_full)
```

[39]:

	deviance	df	deviance_diff	df_diff	F	pvalue
0	109043.809955	2991.000589	NaN	NaN	NaN	NaN
1	107295.111571	2990.000704	1748.698385	0.999884	49.737638	0.000009
2	105030.684668	2987.007254	2264.426903	2.993450	21.513266	0.000026

[40]:

```
gam_full.summary()
```

```
LinearGAM
=============================================
============================================================
Distribution:                        NormalDist Effective DoF:
12.9927
Link Function:                     IdentityLink Log Likelihood:
-14930.261
Number of Samples:                         3000 AIC:
29888.5074
                                                AICc:
29888.648
                                                GCV:
1246.1129
                                                Scale:
35.1625
                                                Pseudo R-Squared:
0.2928
==============================================================================
==========================
Feature Function                  Lambda               Rank         EDoF
P > x        Sig. Code
================================= ==================== ============ ============
============ ============
s(0)                              [465.0491]           20           5.1
1.11e-16     ***
s(1)                              [2.1564]             7            4.0
8.10e-03     **
```

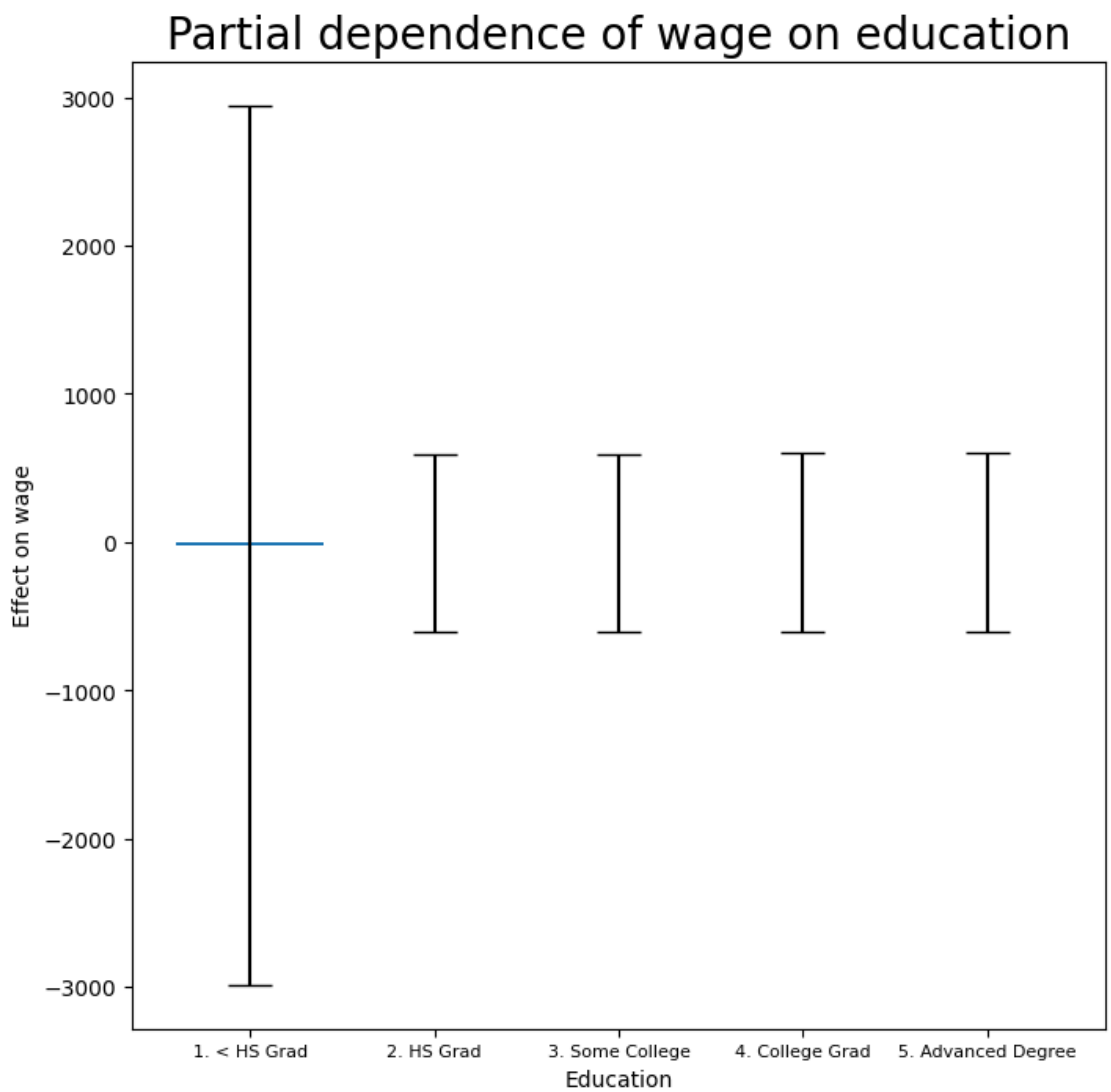
```
f(2)                                [0]                  5            4.0
1.11e-16     ***
intercept                                                1            0.0
1.11e-16     ***
================================================================================
==========================
Significance codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

WARNING: Fitting splines and a linear function to a feature introduces a model
identifiability problem
         which can cause p-values to appear significant when they are not.

WARNING: p-values calculated in this manner behave correctly for un-penalized
models or models with
         known smoothing parameters, but when smoothing parameters have been
estimated, the p-values
         are typically lower than they should be, meaning that the tests reject
the null too readily.

C:\Users\archa\AppData\Local\Temp\ipykernel_30340\2135516388.py:1: UserWarning:
KNOWN BUG: p-values computed in this summary are likely much smaller than they
should be.

Please do not make inferences based on these values!

Collaborate on a solution, and stay up to date at:
github.com/dswah/pyGAM/issues/163

  gam_full.summary()
```

[41]:
```
Yhat = gam_full.predict(Xgam)
```

[42]:
```
gam_logit = LogisticGAM(age_term +
                        l_gam(1, lam=0) +
                        f_gam(2, lam=0))
gam_logit.fit(Xgam, high_earn)
```

[42]:
```
LogisticGAM(callbacks=[Deviance(), Diffs(), Accuracy()],
   fit_intercept=True, max_iter=100,
   terms=s(0) + l(1) + f(2) + intercept, tol=0.0001, verbose=False)
```

[43]:
```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit, 2)
ax.set_xlabel('Education')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of wage on education',
             fontsize=20);
ax.set_xticklabels(Wage['education'].cat.categories, fontsize=8);
```



[44]:
```
pd.crosstab(Wage['high_earn'], Wage['education'])
```

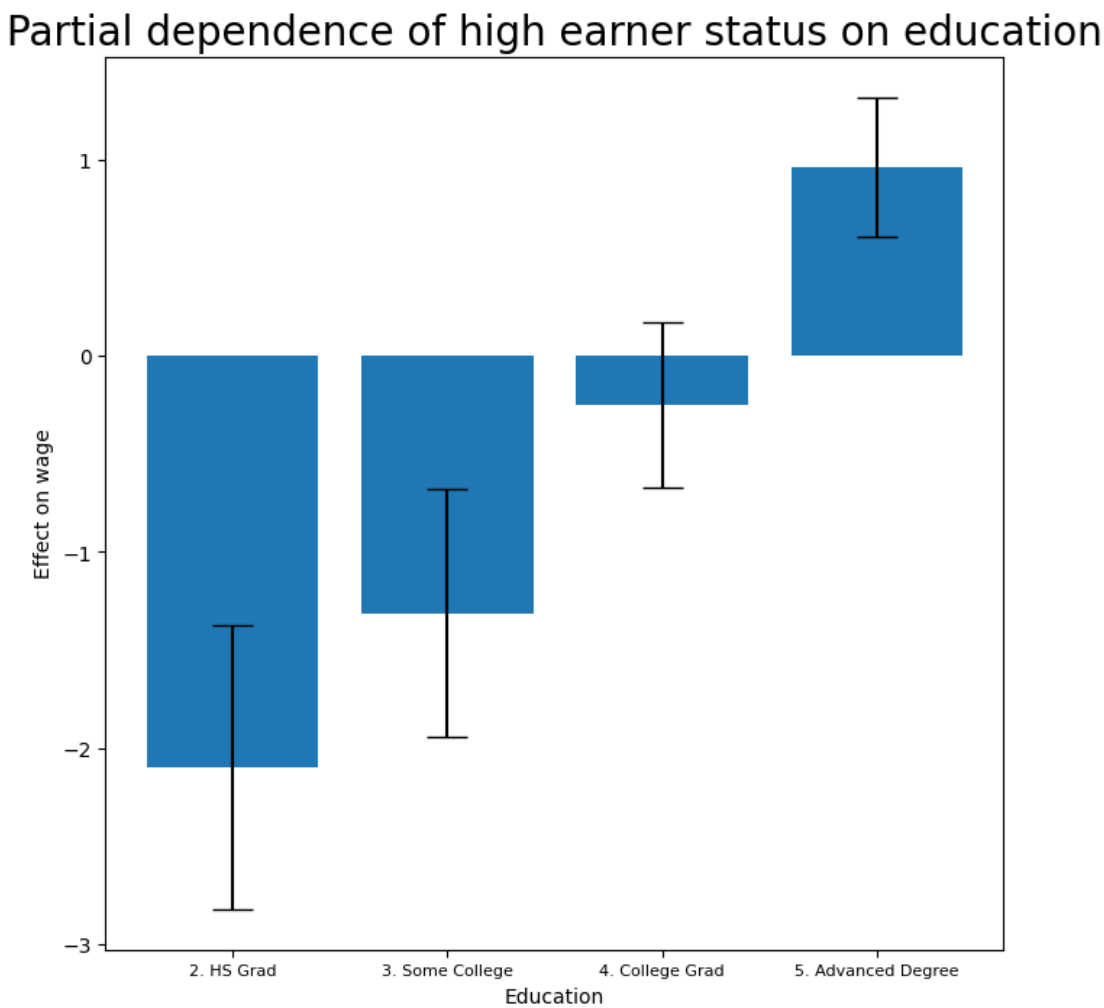
[44]:
```
education  1. < HS Grad  2. HS Grad  3. Some College  4. College Grad  \
high_earn
False               268         966              643              663
True                  0           5                7               22

education  5. Advanced Degree
high_earn
False                    381
True                      45
```

[45]:
```
only_hs = Wage['education'] == '1. < HS Grad'
Wage_ = Wage.loc[~only_hs]
Xgam_ = np.column_stack([Wage_['age'],
                         Wage_['year'],
                         Wage_['education'].cat.codes-1])
high_earn_ = Wage_['high_earn']
```

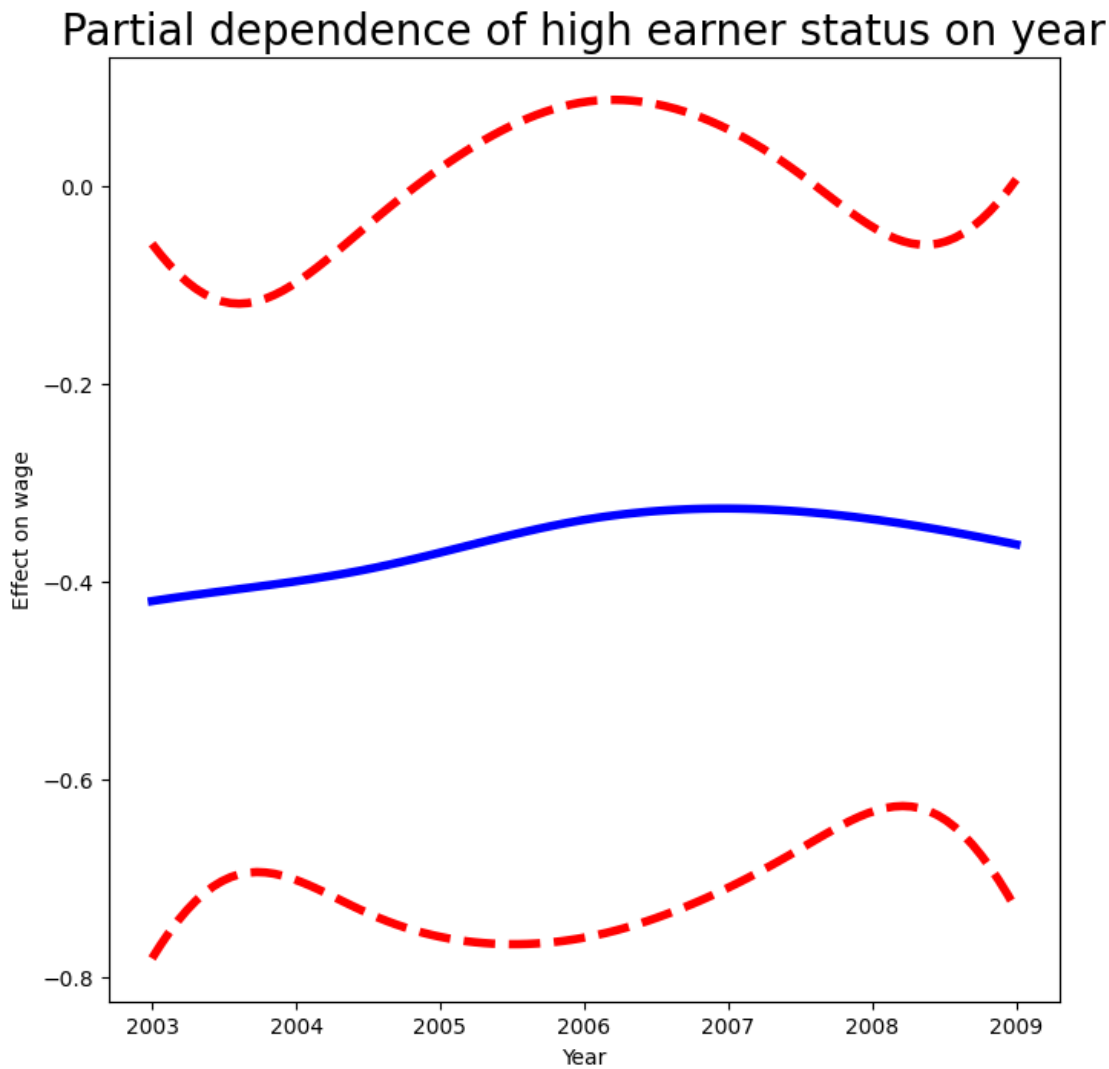
[46]:
```
gam_logit_ = LogisticGAM(age_term +
                         year_term +
                         f_gam(2, lam=0))
gam_logit_.fit(Xgam_, high_earn_)
```

[46]:
```
LogisticGAM(callbacks=[Deviance(), Diffs(), Accuracy()],
   fit_intercept=True, max_iter=100,
   terms=s(0) + s(1) + f(2) + intercept, tol=0.0001, verbose=False)
```

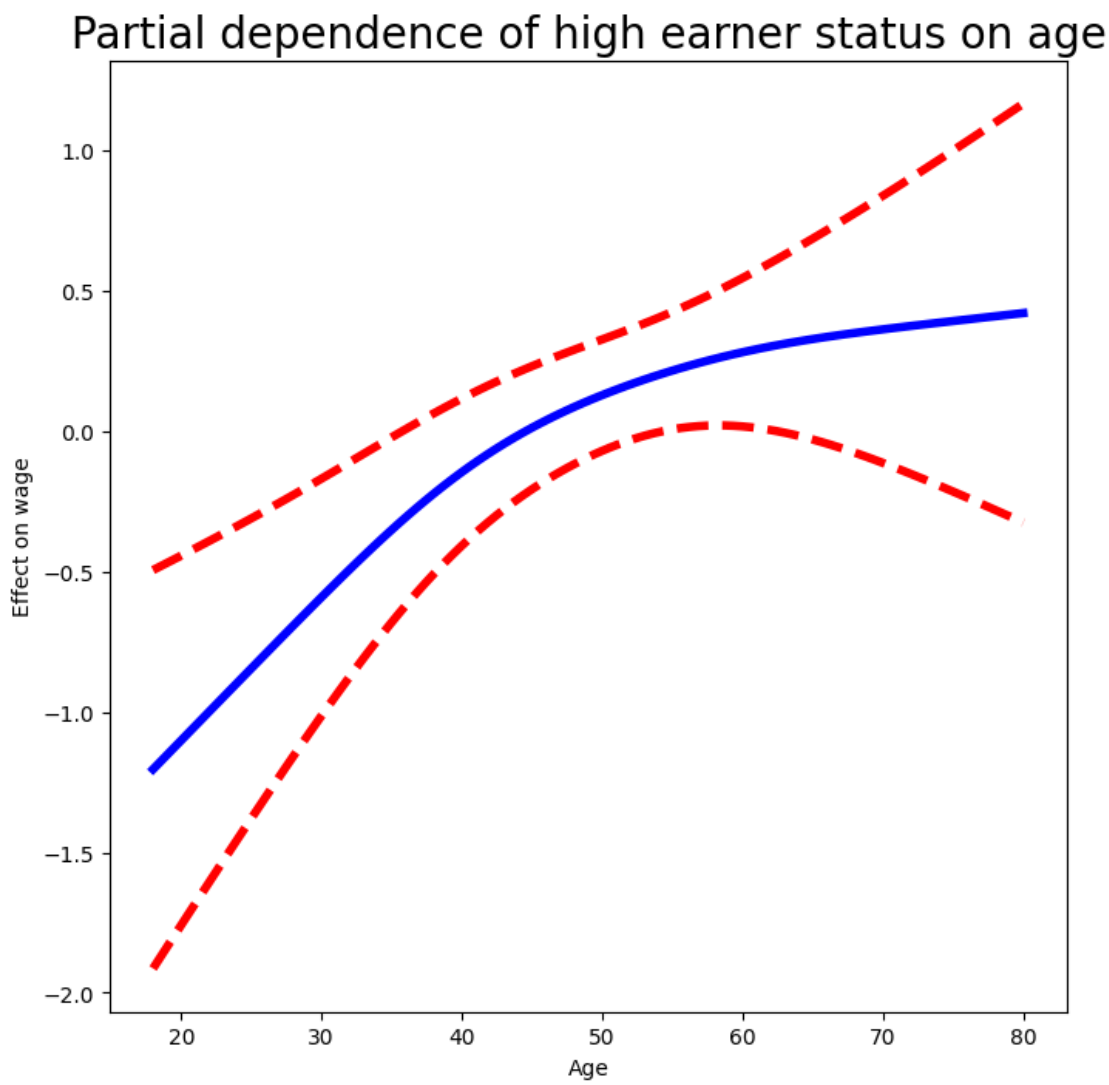
[47]:
```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit_, 2)
ax.set_xlabel('Education')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of high earner status on education',␣
 ↪fontsize=20);
ax.set_xticklabels(Wage['education'].cat.categories[1:],
                   fontsize=8);
```

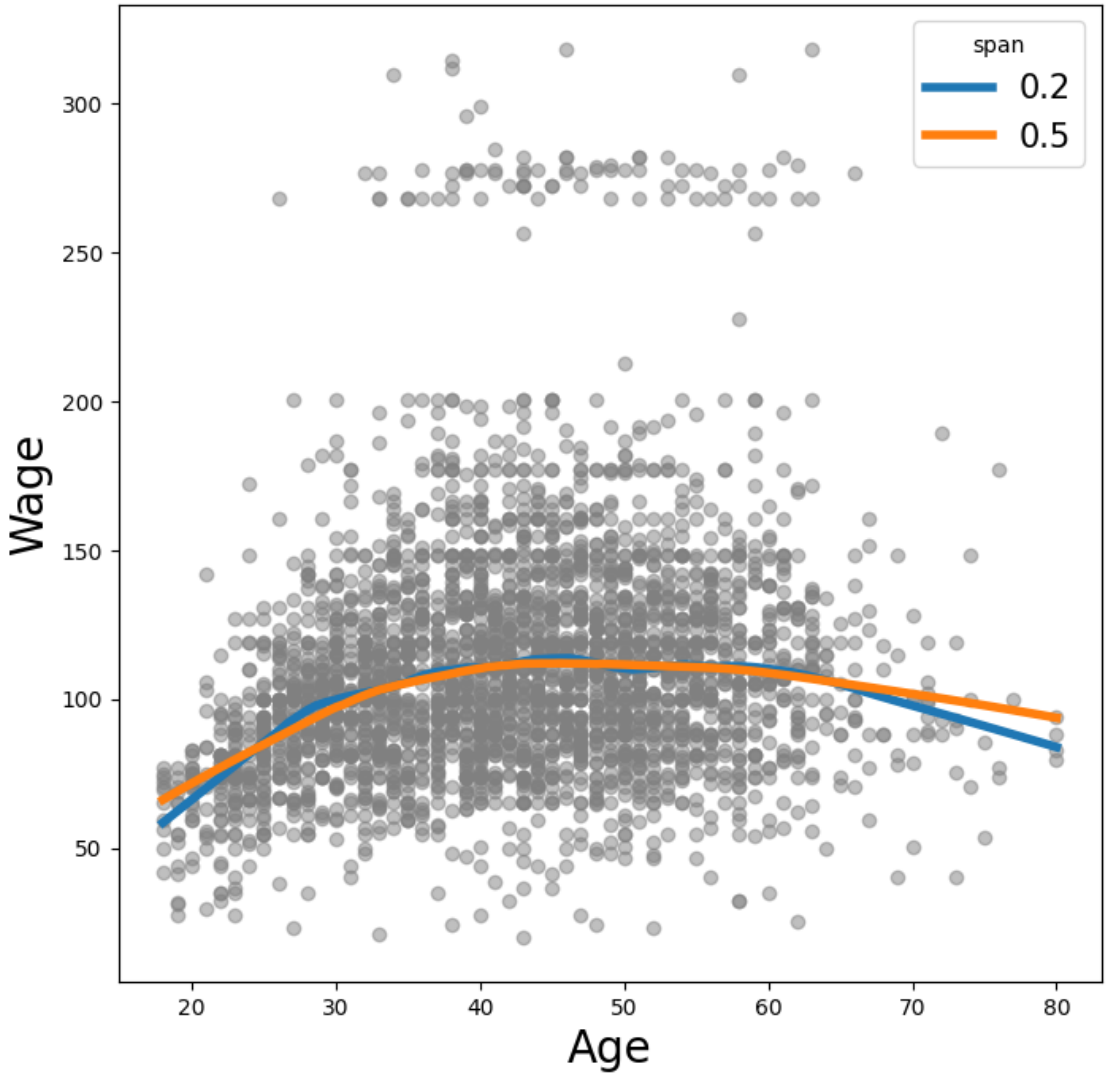


[48]:
```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit_, 1)
ax.set_xlabel('Year')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of high earner status on year',
             fontsize=20);
```



[49]:
```
fig, ax = subplots(figsize=(8, 8))
ax = plot_gam(gam_logit_, 0)
ax.set_xlabel('Age')
ax.set_ylabel('Effect on wage')
ax.set_title('Partial dependence of high earner status on age', fontsize=20);
```



[50]:
```
lowess = sm.nonparametric.lowess
fig, ax = subplots(figsize=(8,8))
ax.scatter(age, y, facecolor='gray', alpha=0.5)
for span in [0.2, 0.5]:
    fitted = lowess(y,
                    age,
                    frac=span,
                    xvals=age_grid)
    ax.plot(age_grid,
            fitted,
            label='{:.1f}'.format(span),
            linewidth=4)
ax.set_xlabel('Age', fontsize=20)
```

```
ax.set_ylabel('Wage', fontsize=20);
ax.legend(title='span', fontsize=15);
```